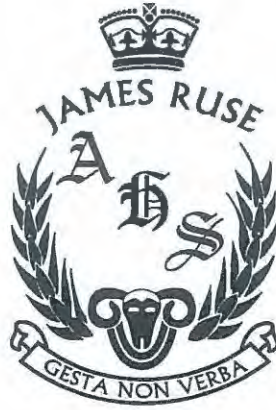


Number:	
Class:	



YEAR 12

ASSESSMENT TEST 3

TERM 2, 2014

MATHEMATICS

EXTENSION 1

Time Allowed – 90 Minutes
(Plus 5 minutes Reading Time)

- All questions may be attempted
- All questions are of equal value
- Department of Education approved calculators are permitted
- In every Question, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged work
- No grid paper is to be used unless provided with the examination paper
- This is an open book test, any printed or hand written materials are allowed must be placed on desk

The answers to all questions are to be returned in separate bundles clearly labeled Question 1, Question 2, etc. Each question must show your Candidate Number.

ATTEMPT ALL 7 QUESTIONS (70 marks)

QUESTION 1 (10 Marks)

COMMENCE A NEW PAGE

MARKS

- a) Five cards numbered 1, 2, 3, 4, 5 are arranged at random.
- (i) How many arrangements are there? 1
 - (ii) What is the probability that the arrangement finishes with a 24? 2
 - (iii) What is the probability that the last number in the arrangement is odd? 2
- (b) The equation of motion of a particle moving along the x -axis is $x = 2 \sin(t + \frac{\pi}{6})$, where x is in metres and t in seconds.
- (i) Draw the displacement-time graph for $0 \leq t \leq \frac{11\pi}{6}$. 2
 - (ii) When does the particle first change direction and where is it at this time? 2
 - (iii) Find the distance travelled by the particle in the first $\frac{4\pi}{3}$ seconds. 1

QUESTION 2 (10 Marks)

COMMENCE A NEW PAGE

- (a) A particle moves in a straight line so that its acceleration as a function of displacement is given by $\frac{d^2x}{dt^2} = 1 - 4x$. Initially $x = 1.25$ cm and $v = 0$.
- (i) Show that the motion is simple harmonic. 1
 - (ii) Find the amplitude, period and centre of motion. 3
 - (iii) Find the velocity when $t = 0.2$ seconds. 2
 - (iv) Describe briefly what would have happened if the motion had commenced at $x = 0.25$ with $v = 0$. 1
- (b) A container with capacity A litres is being filled with water. After t minutes the volume V litres of water in the container is given by $V = A(1 - e^{-kt})$ for some constant $k > 0$. 3
- If one quarter of the container is filled in the first two minutes, find what fraction of the container is filled in the next two minutes.

QUESTION 3 (10 Marks)**COMMENCE A NEW PAGE****MARKS**

- (a) During a chemical reaction the amount, A grams, of a substance unconverted after t hours is given by the formula $A = 4 + e^{-3kt}$.
- (i) Show that $\frac{dA}{dt} = -3k(A - 4)$. 1
- (ii) If initially A decreases at the rate of 0.04 grams per hour, find the value of k . 2
- (iii) Using k found in part (ii), sketch the graph of $A = 4 + e^{-3kt}$ and indicate the values that A can take. 3
- (b) Eight people attend a meeting. They are provided with two circular tables, one seating 3 people, the other 5 people.
- (i) How many seating arrangements are possible? 2
- (ii) If the seating is done randomly, what is the probability that a particular couple are on different tables? 2

QUESTION 4 (10 Marks)**COMMENCE A NEW PAGE**

- (a) A container of water, heated to 100°C , is placed in a coolroom where the temperature is maintained at -5°C . After t minutes, the rate of change of the temperature, $T^{\circ}\text{C}$, of the water is given by $\frac{dT}{dt} = -k(T + 5)$, where k is a constant.
- (i) Assuming the function $T = Ae^{-kt} - 5$, where A is a constant, is a solution to the above differential equation, find the value of A . 1
- (ii) After 20 minutes, the water temperature falls to 30°C . Find, to the nearest degree, the water temperature after a further 10 minutes. 2
- (iii) Find, to the nearest minute, the time the water will need to be in the coolroom before its temperature reaches 0°C . 2
- (b) A small cube has one red face, two blue faces and three green faces. It is rolled three times.
- (i) Find the number of distinct probability values of the possible outcomes. 1
- (ii) Find the probability that only one colour appeared in three rolls of the cube. 2
- (iii) Find the probability that exactly one blue face appeared in three rolls of the cube. 2

QUESTION 5 (10 Marks)**COMMENCE A NEW PAGE****MARKS**

- (a) Two bags each contain three marbles, one red and two blue. A marble is drawn at random from the first bag and placed in the second. A marble is drawn from the second and placed in the first. 2

Find the probability that each bag still contains one red and two blue marbles.

- (b) The letters of the word PERSEVERE are randomly arranged in a row. Find the probability that the E's are together and the R's are together? 2

- (c) A particle vibrates in Simple Harmonic Motion, making 100 oscillations per second.

- (i) Show that the acceleration is given by $\ddot{x} = -40\,000\pi^2 x$. 2

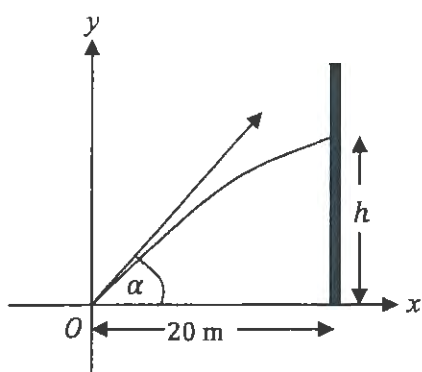
- (ii) If the amplitude of the motion is 20 cm, calculate the speed of the particle at

- α) the centre of its motion. 3

- β) the extremities of its motion. 1

QUESTION 6 (10 Marks)**COMMENCE A NEW PAGE**

- (a) A garden hose releases a stream of water with a velocity of 20 m s^{-1} at an angle of α to the horizontal. The water streams toward a high wall 20 m away on level ground.



Given $x = 20t \cos \alpha$ and $y = 20t \sin \alpha - 5t^2$,
where x and y are the horizontal and vertical displacements of the stream of water from O at any time t , $g = 10 \text{ m s}^{-2}$ and the coordinate axes are taken as shown.

- (i) Find the equation of the path of the stream of water in terms of x , y and α . 1

- (ii) Show that the height h at which the water hits the wall is given by 2
$$h = 20 \tan \alpha - 5(1 + \tan^2 \alpha).$$

- (iii) Using part (ii), show that the maximum value of h occurs when 2
 $\tan \alpha = 2$ and find this maximum height.

- (b) A particle moves in a straight line so that after t seconds its velocity is $v \text{ m s}^{-1}$ and its displacement is x .

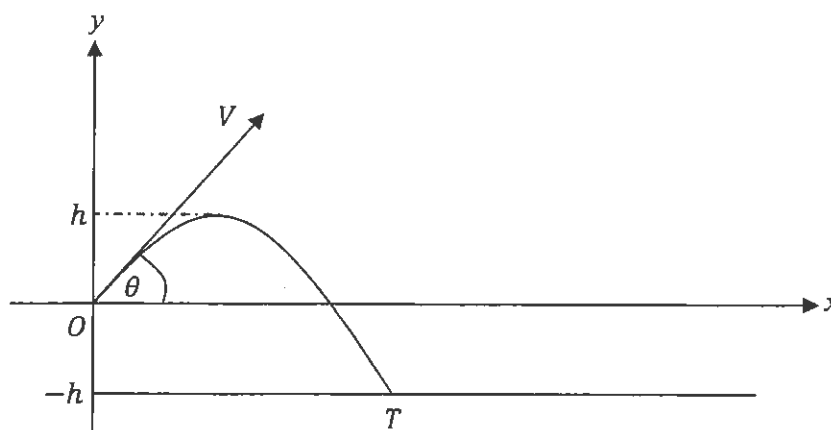
(i) Given that $\frac{d^2x}{dt^2} = 10x - 2x^3$ and that $v = 0$ when $x = -1$, find v in terms of x . 3

- (ii) Explain why the motion cannot exist between $x = -1$ and $x = 1$. 2

QUESTION 7 (10 Marks)

COMMENCE A NEW PAGE

The diagram below shows the path of a projectile fired from the top of a cliff, O . Its initial velocity is $V \text{ m s}^{-1}$, its initial angle of elevation is θ and it rises to a maximum height h metres above O . It strikes a target T situated on a horizontal plane h metres below O .



- a) Given that $\ddot{y} = -g$ and $\ddot{x} = 0$, derive equations for y and x as functions of time. 2

- b) Prove that $h = \frac{V^2 \sin^2 \theta}{2g}$, where g is acceleration due to gravity. 3

- c) Prove that the time taken for the projectile to reach its target is $\frac{V(1 + \sqrt{2}) \sin \theta}{g}$ seconds. 3

- d) Show that the distance from the target to the base of the cliff is $\frac{V^2(1 + \sqrt{2}) \sin 2\theta}{2g}$ metres. 2

☺ END OF EXAMINATION ☺

Suggested Solutions	Marks	Marker's Comments
a) i) First card chosen in 5 ways Second " " " 4 ways " " " " " 3 ways Last " " " 1 way <u>5! ways</u> <u>= 120 ways</u> ii) Last 2 cards already placed Others can be placed in $3! = 6$ ways $\therefore p(E) = \frac{3!}{5!} = \frac{6}{120} = \frac{1}{20}$ iii) Last number is odd in 3 ways out of 5. $\therefore p(E) = \frac{3}{5}$	1 1 1 2	5! was <u>not</u> enough. Some people found 72 ways of doing this
b) i) ii) From the graph, first direction change is when <u>$t = \pi/3$ sec.</u> and <u>$x = 2$ m.</u> iii) Between 0 and $\pi/3$ travels 1 (+ve) $\pi/3$ and $2\pi/3$ travels 4 (-ve) <u>total distance = 5 metres</u>	2 1 1 1	This diagram shows the lowest minimum for 2 marks. Any omissions of error lost 1 mark.

MATHEMATICS Extension 1 : Question.....2

Suggested Solutions	Marks	Marker's Comments
a) (i) $\frac{d^2x}{dt^2} = 1 - 4x$ $= -4\left(x - \frac{1}{4}\right)$ $= -2^2\left(x - \frac{1}{4}\right)$ Since of form $\ddot{x} = -n^2 X$ where $X = (x - x_0)$ it is in SHM (ii) From (i) centre is $\frac{1}{4}$ (ii) period = $\frac{2\pi}{2} \left(\frac{2\pi}{n}\right)$ $= \pi$ amplitude since extremity is 1.25 and centre 0.25 amplitude = $1.25 - 0.25$ $= 1$ Alternative not as elegant $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = 1 - 4x$ $\frac{1}{2}v^2 = x - 2x^2 + C_1$ $v^2 = 2x - 4x^2 + C_2$ at $x = 1.25$ $v = 0$ $0 = 2 \times (1.25) - 4 \times (1.25)^2 + C_2$ $\therefore \underline{C_2 = 3.75}$	1 1 1 1	need to make a clear statement as to why it is in SHM. Too many students used $v^2 = n^2(a^2 - x^2)$ which only works when centre is origin

MATHEMATICS Extension 1 : Question...2

Suggested Solutions	Marks	Marker's Comments
$v^2 = 2x - 4x^2 + 3.75$ $v = 0$ $0 = 4x^2 - 2x - 3.75$ $0 = 16x^2 - 8x - 15$ $(4x + 3)(4x - 5) = 0$ $x = -\frac{3}{4} \text{ or } \frac{5}{4}$ $\therefore \text{amplitude } \frac{1}{2} \times \left(\frac{5}{4} - -\frac{3}{4} \right)$ $= 1$ (iii). Assume general equation $x = b + a \cos(nt + \alpha)$ $b = \frac{1}{4} \text{ when } t = 0 \quad x = 1.25$ $a = 1$ $n = 2$ $1.25 = \frac{1}{4} + \cos(2t + \alpha)$ $1 = \cos \alpha$ $\therefore \alpha = 0$ $x = \frac{1}{4} + \cos(2t)$ OR $(x = \frac{1}{4} + \sin(2t + \pi/2))$	1	Students using this method made algebraic errors. Some students found this but substituted in $v^2 = n^2(a^2 - x^2)$ rather than just differentiating

MATHEMATICS Extension 1: Question 2

Suggested Solutions	Marks	Marker's Comments
$\frac{dx}{dt} = -2 \sin 2t$ when $t = 0.2$ $v = -2 \sin(0.4)$ $= -0.78 \text{ m/s.}$ (b) $v = A - Ae^{-kt}$ $\frac{A}{4} = A - Ae^{-2k}$ $\frac{1}{4} = 1 - e^{-2k}$ $-2k = \ln\left(\frac{3}{4}\right)$ $k = \frac{1}{2} \ln\left(\frac{4}{3}\right)$ $v = A\left(1 - e^{-\frac{1}{2} \ln\left(\frac{4}{3}\right)t}\right)$ $v = A\left(1 - e^{-2 \ln\left(\frac{4}{3}\right)}\right)$ $v = A\left(1 - e^{\ln \frac{9}{16}}\right)$ $= A\left(1 - \frac{9}{16}\right)$ $= \frac{7}{16} A$ In next 2 minutes $\frac{7}{16} A - \frac{1}{4} A = \frac{3}{16} A$ so $\frac{3}{16}$ will be filled in next 2 minutes	1 1 1	• need to evaluate $\sin(0.4)$ in radians - • students substituted $t = 0.2$ into $v = -2 \sin(2t)$ Some students used $v = \frac{1}{4}$ OR $v = \frac{v}{4}$ which made question difficult. Many students failed to answer question asking

3-unit Mathematics

Question 3

(i) show that $\frac{dA}{dt} = -3k(A-4)$

$$A = 4 + e^{-3kt}$$

$$\frac{dA}{dt} = -3k e^{-3kt}$$

$$\text{since } A = 4 + e^{-3kt}$$

$$e^{-3kt} = A - 4$$

$$\therefore \frac{dA}{dt} = -3k(A-4)$$

Alternatively

$$\begin{aligned} \frac{dA}{dt} &= -3k(e^{-3kt}) \\ &= -3k(4 + e^{-3kt} - 4) \\ &= -3k(A-4) \end{aligned}$$

(ii) Find the value of k if initially A decreases at a rate of 0.04 g/hr.

when $t=0$, $\frac{dA}{dt} = -0.04$.

$$\frac{dA}{dt} = -3k e^{-3k(0)}$$

$$-0.04 = -3k e^0$$

$$-0.04 = -3k$$

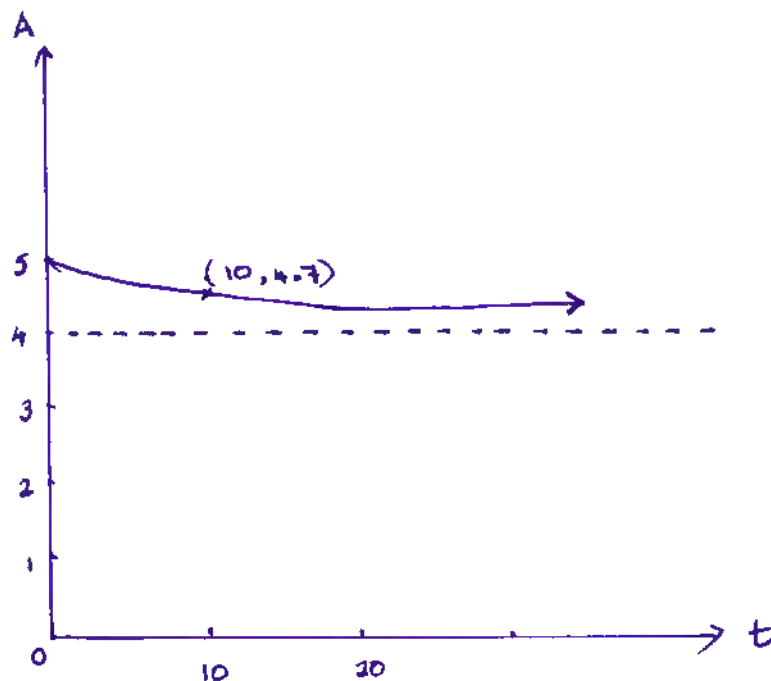
$$\therefore k = \frac{-0.04}{-3}$$

$$k = \frac{1}{75}$$

① If used $\frac{dA}{dt}$ as 0.04 then got $-\frac{1}{75}$ as answer (got only 1 mark).

② If used $\frac{dA}{dt} = -3k(A-4)$ then had to calculate A as 5 .

(iii)



1 mark for y-intercept
1 mark for asymptote $y=4$
1 mark for shape and must show one other point apart from y-intercept

Note

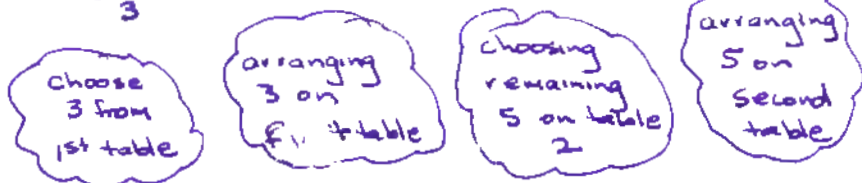
Took away marks if curve was touching the asymptote.

Other points mentioned were $(3, 4.84)$, $(200, 4.0)$
 $(5, 4.82)$, $(7.3, 4.5)$ - accepted $(25, 4 + \frac{1}{e})$

b) (i) Number of seating arrangements possible



$${}^8C_3 \times (3-1)! \times {}^5C_5 \times (5-1)!$$



$$= \frac{8!}{5!3!} \times 2! \times 1 \times 4!$$

$$= 2688 \text{ ways}$$

$${}^8C_5 \times 4! \times {}^3C_3 \times 2! = 2688 \text{ ways}$$

(2)

2

Some did

- Split 3 people and 5 people as

$${}^8C_3 \times {}^5C_5 = 56$$

and arranging around table as

$$2! \times 4! = 48$$

$$\therefore \text{Total ways} = 56 \times 48 = 2688$$

✓ 1 mark for choosing 3 or 5 on a table
ie ${}^8C_3 \times {}^5C_5$

$$\text{or } {}^8C_5 \times {}^3C_3$$

✓ 1 mark arranging 3 or 5 on a table
ie $2! \times 4!$

Some got:

$$* {}^8C_3 \times 2! \times {}^5C_4 \times 4! = 13440$$

No need as choosing 5 people on 2nd table is fixed

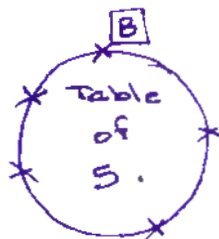
$$* 40320 \text{ means you did } {}^8C_5 \times {}^3C_3 \times 3! \times 5!$$

$$* 5040 = 7!$$

$$* 8! = 40320$$

b(ii) P(particular couple are on different tables)

Say A and B are couples



2 are already seated i.e. A and B
Now we have 6 to choose from.

$${}^6C_2 \times (3-1)! \times (5-1)! \times 2$$

select
2 more
to sit
on 1st
table

arranging
seating
small
table

arranging
seating on
large
table

swap
couple

$$= 1440 \text{ ways.}$$

$$\therefore P(\text{couple on different tables}) = \frac{1440}{2688} = \frac{15}{28}$$

Method II

$$P(\text{couple on different tables}) = 1 - P(\text{couple on same table})$$

$$= 1 - \left({}^6C_3 \times {}^3C_2 \times 4! \times 2! + {}^6C_5 \times {}^1C_1 \times 4! \times 2! \right) \div 2688$$

Couple together
on table of 5

Couple
together
on table
of 3

$$= 1 - \frac{1248}{2688}$$

$$= 1 - \frac{13}{28}$$

$$= \frac{15}{28}$$

2

Different way 5.

3

A on table 3

$${}^6C_2 \times 2! \times 4! = 720$$

A on table of 5

$${}^6C_4 \times 2! \times 4! = 720$$

Case 1 A on table 3
and B on table 5

$${}^6C_2 \times 2! \times {}^4C_4 \times 4!$$

Case 2 B on table of 3
and A on table of 5

$${}^6C_2 \times 2! \times {}^4C_4 \times 4!$$

$$2C_1 \times {}^6C_2 \times 2! \times 4!$$

choose
which
table
each
person
will
sit in

choose
2
on
table
of
3

ways
of
arranging
on
table 3

ways
of
arranging
on
table 5

Note:

if got $\frac{15}{56}$ means

you did not
swap the
seating i.e. $\times 2$

Method III

Case 1
Couple on same table of 3



$${}^6C_1 \times 2! \times {}^5C_5 \times 4! = 288$$

Case 2 couple on same table of 5.

$${}^6C_3 \times 4! \times {}^3C_3 \times 2! = 960$$

$$\begin{aligned} \therefore p(\text{couple on different tables}) &= 1 - p(\text{couple on same table}) \\ &= 1 - \frac{1248}{2688} \\ &= \frac{1440}{2688} \\ &= \frac{15}{28} \end{aligned}$$

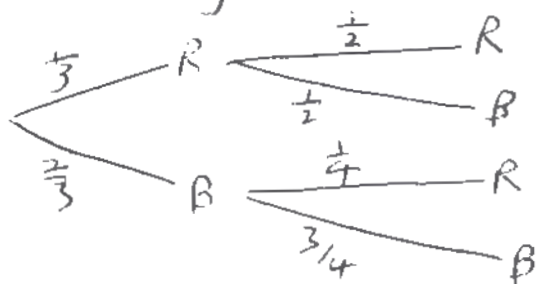
30 MATHEMATICS: Question 4..

Suggested Solutions	Marks	Marker's Comments
(i) (1) $T = Ae^{-kt} - 5$ When $T = 100$ $t = 0$ $100 = Ae^0 - 5$ $\therefore A = 105$ (ii) When $t = 20$ $T = 30$ $30 = 105e^{-20k} - 5$ $35 = 105e^{-20k}$ $-20k = \ln \frac{35}{105}$ $k = -\frac{1}{20} \ln \frac{35}{105}$ $= -\frac{1}{20} \ln \frac{1}{3}$ OR $\frac{\ln 3}{20}$ When $t = 30$ $T = 105e^{-30k} - 5$ $= 105e^{-30(\frac{\ln 3}{20})} - 5$ $= 15 \cdot 20725942$ $= 15^\circ$ (nearest degree) (iii) When $T = 0$ $0 = 105e^{-kt} - 5$ $e^{-kt} = \frac{5}{105}$ $e^{-kt} = \frac{1}{21}$ $-kt = \ln \frac{1}{21}$ $t = -\frac{1}{k} \ln \frac{1}{21}$ OR $\frac{\ln 21}{k}$ $t = \frac{\ln \frac{1}{21}}{-\frac{1}{20} \ln \frac{1}{3}}$ OR $\frac{20 \ln 21}{\ln 3}$ $= 55.42487498$ $= 55$ min (nearest min)	1 mk 1 mk 1 mk 1 mk	Two common careless errors (1) Forget the -5 (2) Subtracted 5 from 100 & got $A = 95$. Many students confused t and T in the substitution Common error - forgot to subtract the 5 Mark not deducted for not rounding. Note: w/o $t = 55$ $T = 0.12$ $t = 56$ $T = 0.56$
However because the temperature is dropping it would not pass 0° until 56 min.		

30 MATHEMATICS: Question 4

Suggested Solutions	Marks	Marker's Comments
(b)(i) $RRR = \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6}$, $R RB = \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{3}$, $RRG = \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{2}$ $BBR = \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{6}$, $BBB = \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3}$, $B BG = \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{2}$ $GGG = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{6}$, $GGB = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{3}$, $GGG = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$ $R BG = \frac{1}{6} \cdot \frac{1}{3} \cdot \frac{1}{2}$ There are <u>10</u> distinct probability values		A common incorrect answer was 27. Note: There are 27 different combinations but not distinct, eg. $BBR = RBB = \frac{1}{54}$ $GGB = GBG = \frac{1}{12}$ etc.
(ii) $P(RRR) + P(BBB) + P(GGG)$ $= \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$ $= \left(\frac{1}{6}\right)^3 + \left(\frac{1}{3}\right)^3 + \left(\frac{1}{2}\right)^3$ $= \frac{1}{216} + \frac{1}{27} + \frac{1}{8}$ $= \frac{1}{6}$	1mk 1mk	
(iii) $P(\text{exactly one blue}) = P(B\bar{B}\bar{B}) + P(\bar{B}B\bar{B}) + P(\bar{B}\bar{B}B)$ $= \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{2}{3} + \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{2}{3} + \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{1}{3}$ $= 3 \left(\frac{1}{3} \cdot \frac{2}{3} \cdot \frac{2}{3} \right)$ $= \frac{4}{9}$	1mk	Students who drew a tree diagram had a better understanding of the question Note: If you were looking at all colours ③ RRB, RBR, BRR ③ GGB, GBG, BGG ③ RRG, RGR, BRR, BGR, GRR, GBR

a) 1st bag 2nd bag



$$\begin{aligned}
 P(E) &= P(RR) + P(BB) \\
 &= \frac{1}{3} \times \frac{1}{2} + \frac{2}{3} \times \frac{3}{4} \\
 &= \frac{1}{6} + \frac{1}{2} \\
 &= \frac{2}{3} \#
 \end{aligned}$$

1 m for $\frac{1}{6}$ or $\frac{1}{2}$
1 m for $\frac{2}{3}$

well

b) PERSEVERE (4 E) (2 R) (1 P) (1 V) (1 S)
9 letters

done

No of arrangements with E's together & R's together
or $5! = 120$

No of arrangements w/o restriction = $\frac{9!}{4!2!} = 7560$
(4 E) (2 R)

$$P(E) = \frac{120}{7560} = \frac{1}{63} \#$$

1 m for $5!$ or $\frac{9!}{4!2!}$

1 m for $\frac{1}{63}$

$$\begin{aligned}
 c) \quad T &= \frac{2\pi}{\omega} = \frac{1}{100} \quad \therefore \omega = 200\pi \\
 \ddot{x} &= -\omega^2 x = -40,000\pi^2 x
 \end{aligned}$$

1 m for ω
1 m for sub.

C.O.M 1 m
 $x=0$

$$\begin{aligned}
 \therefore) \quad \alpha) \quad \text{centre of motion } x &= 0 \\
 v^2 &= \omega^2 (a^2 - x^2) \\
 &= 40000\pi^2 (20^2 - 0) \\
 v^2 &= 40,000\pi^2 \times 400 \\
 v &= \pm 4000\pi \text{ cm/s}
 \end{aligned}$$

1 m

$$\text{speed} = |v| = 4000\pi \text{ cm/s} \# \text{ or } 4\pi \text{ m/s} \#$$

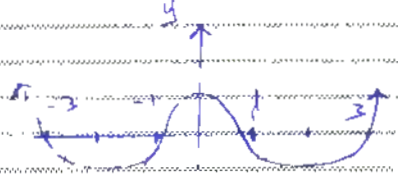
1 m must
justify speed
is positive &
must have
correct unit

$$\begin{aligned}
 \beta) \quad \text{When } x &= \pm 20, \\
 v &= 0 \text{ cm/s} \#
 \end{aligned}$$

EXTENSION I
MATHEMATICS: Question 6

Suggested Solutions	Marks	Marker's Comments
(i) $x = 20t \cos \alpha$ $y = 20t \sin \alpha - 5t^2$ $\therefore t = \frac{x}{20 \cos \alpha}$ $y = 20 \left[\frac{x}{20 \cos \alpha} \right] \sin \alpha - 5 \left[\frac{x}{20 \cos \alpha} \right]^2$ $= x \tan \alpha - \frac{5x^2}{400 \cos^2 \alpha}$ $= x \tan \alpha - \frac{x^2}{80 \cos^2 \alpha}$ (ii) $y = x \tan \alpha - \frac{x^2 \sec^2 \alpha}{80}$ $= x \tan \alpha - \frac{x^2 (1 + \tan^2 \alpha)}{80}$ when $y = h$ $x = 20$ $h = 20 \tan \alpha - \frac{(20)^2 (1 + \tan^2 \alpha)}{80}$ $= 20 \tan \alpha - 50 (1 + \tan^2 \alpha)$ $\therefore h = 20 \tan \alpha - 5 (1 + \tan^2 \alpha)$ (iii) Method (1) Equation for h is a quadratic in $\tan \alpha$. Let $\tan \alpha = t$ $h = 20t - 5(1 + t^2)$ $= -5t^2 + 20t - 5$ As coefficient of $t^2 < 0$ Parabola opens downwards Maximum h occurs when $t = \frac{-b}{2a} = \frac{-20}{2(-5)} = 2$ max x when $\tan \alpha = 2$ Method (2) $\frac{dh}{dt} = -10t + 20$ for stationary $\frac{dh}{dt} = 0$ $t = 2$ $\frac{d^2h}{dt^2} = -10 < 0$ concave down local max = absolute max $\tan \alpha = 2$	① ② ②	Correct Equation in any form involving x, y, α. NOT t ① for changing $\frac{1}{\cos^2 \alpha}$ to $(1 + \tan^2 \alpha)$ ① sub values and simplifying Answer was given so substitution must be shown ① for showing max occurs when $\tan \alpha = 2$ must show why it is maximum Alternatively find $\frac{dh}{d\alpha}$ and test formula

EXTENSION I
MATHEMATICS: Question 6

Suggested Solutions	Marks	Marker's Comments
max number of times $x = 2$ $h = 20 \times 2 - 5(1 + 2^2)$ $= 40 - 25$ $= 15$ max height is 15 m b) (i) $\frac{d^2x}{dt^2} = 10x - 2x^3$ $\frac{d(\frac{1}{2}v^2)}{dx} = 10x - 2x^3$ $\frac{1}{2}v^2 = \frac{10x^2}{2} - \frac{2x^4}{4} + C$ $v = 0 \quad x = 1$ $0 = 5 - \frac{1}{2} + C$ $C = -9/2$ $\therefore \frac{1}{2}v^2 = 5x^2 - \frac{1}{2}x^4 - 9$ $v^2 = 10x^2 - x^4 - 9$ $\therefore v = \pm \sqrt{10x^2 - x^4 - 9}$ (ii) For motion to exist $v^2 > 0$ Method 1 $\therefore 10x^2 - x^4 - 9 > 0$ $x^4 - 10x^2 + 9 < 0$ $\therefore \text{consider } y = x^4 - 10x^2 + 9$ $y = (x^2 - 9)(x^2 - 1)$ $= (x - 3)(x + 3)(x - 1)(x + 1)$  $\therefore x^4 - 10x^2 + 9 > 0$ for $-1 < x < 1$ $v^2 < 0 \text{ for } -1 < x < 1 \therefore \text{No motion}$ Method 2 $v = 0$ at $x = \pm 3$ $x = \pm 1$ at $x = -1$ $v = 0$ $\ddot{x} = -8$ object moves to left and stops at $x = -1$ at $x = -3$ $\ddot{x} = 24$ object moves to right And stops at $x = -3$ then moves to left oscillating between $x = -3$ and $x = -1$ $\therefore$ No motion for $-1 < x < 1$	① ③ ②	① correct answer using $\frac{d}{dx}(\frac{1}{2}v^2)$ ① integrating ① C value ① complete expression including $\pm \sqrt{\quad}$ ① stating $v^2 > 0$ ① showing that $v^2 < 0$ for $-1 < x < 1$ Alternatively: ① showing particle moves to left then returning ① showing particle only moves between $-3 < x < -1$

1/2

MATHEMATICS Extension 1 : Question...

Suggested Solutions	Marks	Marker's Comments
(a) $\ddot{x} = 0$ $\dot{x} = c_1$ At $t = 0, x = 0, \dot{x} = v \cos \theta$ $\therefore c_1 = v \cos \theta$ $\therefore \dot{x} = v \cos \theta$ $x = vt \cos \theta + c_3$ At $t = 0, x = 0, y = 0 \therefore c_3 = 0$ and $c_4 = 0$ $\therefore x = vt \cos \theta$ $\ddot{y} = -g$ $\dot{y} = -gt + c_2$ $\dot{y} = v \sin \theta$ $c_2 = v \sin \theta$ $\therefore \dot{y} = -gt + v \sin \theta$ $y = -\frac{g}{2}t^2 + vt \sin \theta + c_4$ At $t = 0, x = 0, y = 0 \therefore c_3 = 0$ and $c_4 = 0$ $\therefore y = -\frac{g}{2}t^2 + vt \sin \theta$		* If the student didn't show how they evaluated the constants of integration, they get 0. ① for 'x' ① for 'y'
(b) max height when $\dot{y} = 0$ $\therefore 0 = -gt + v \sin \theta$ $t = \frac{v \sin \theta}{g}$ sub into y $\therefore h = -\frac{g}{2} \left(\frac{v \sin \theta}{g} \right)^2 + v \left(\frac{v \sin \theta}{g} \right) \sin \theta$ $= -\frac{v^2 \sin^2 \theta}{2g} + \frac{v^2 \sin^2 \theta}{g}$ $= \frac{-v^2 \sin^2 \theta + 2v^2 \sin^2 \theta}{2g}$ $= \frac{v^2 \sin^2 \theta}{2g}$	① mk ① ①	
(c) when $y = -h$ $-h = -\frac{1}{2}gt^2 + vt \sin \theta$ $-\frac{v^2 \sin^2 \theta}{2g} = -\frac{1}{2}gt^2 + vt \sin \theta$	①	

2/2

MATHEMATICS Extension 1 : Question 7 cont.

Suggested Solutions

Marks

Marker's Comments

$$\therefore 0 = gt^2 - 2vt \sin \theta - \frac{v^2 \sin^2 \theta}{g}$$

$$\therefore t = \frac{2v \sin \theta \pm \sqrt{(-2v \sin \theta)^2 - 4g\left(\frac{-v^2 \sin^2 \theta}{g}\right)}}{2g}$$

$$= \frac{2v \sin \theta \pm \sqrt{4v^2 \sin^2 \theta + 4v^2 \sin^2 \theta}}{2g}$$

$$= \frac{2v \sin \theta \pm 2\sqrt{2}v \sin \theta}{2g}$$

$$= \frac{v \sin \theta \pm \sqrt{2}v \sin \theta}{g}$$

$$= \frac{v \sin \theta (1 + \sqrt{2})}{g}$$

as $t > 0$
time is positive

①

①

had to
explain why
it was positive

(d) when $t = \frac{v \sin \theta (1 + \sqrt{2})}{g}$, $x = ??$

$$\therefore x = vt \cos \theta$$

$$= v \cdot \frac{v \sin \theta (1 + \sqrt{2})}{g} \cos \theta$$

$$= \frac{v^2 \sin \theta \cos \theta (1 + \sqrt{2})}{g}$$

$$= \frac{v^2 2 \sin \theta \cos \theta (1 + \sqrt{2})}{2g}$$

$$= \frac{v^2 \sin 2\theta (1 + \sqrt{2})}{2g}$$

①

①